

Udarata Dance Pose Detection and Validation Using Computer Vision and Deep Neural Networks: A Comparative Study of Static and Dynamic Approaches

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Abstract— Udarata dance, a traditional Sri Lankan art form, requires precise body movements and poses. This research introduces a novel approach for detecting and validating these poses using computer vision and deep neural networks. We explore two methodologies: static pose detection, which compares images against a pre-labeled dataset, and dynamic pose detection, which employs Mediapipe to track body landmarks in videos and an LSTM model to classify sequences of poses as either correct or incorrect. Our findings highlight the effectiveness of these methods in accurately identifying proper Udarata dance poses, offering a valuable tool for dance instruction and cultural preservation. For our experiments, we have defined key angles and heights of body parts and compiled a new Udarata Dance Dataset. The experimental results demonstrate that our algorithms achieve superior performance in pose detection and validation.

Keywords—Pose detection, Computer vision, Deep neural networks, Static pose detection, Dynamic pose detection, Mediapipe, LSTM Model, Body Landmarks, Dance Validation

I. INTRODUCTION

Udarata dance, originating from the central highlands of Sri Lanka, is renowned for its intricate and precise movements. Mastery of these poses is crucial for both performers and instructors. Traditionally, ensuring pose accuracy has relied on the expertise of seasoned dancers, but this approach is subjective and labor-intensive. This research aims to automate the detection and validation of Udarata dance poses using advanced computer vision techniques and deep learning models. By doing so, we seek to provide an objective, scalable solution that can aid in both teaching and preserving this cultural heritage. The paper is structured as follows: we first review relevant literature, then detail our dataset preparation and model development processes, followed by an evaluation of our methods and a discussion of our findings. This classifies Udarata dances by extracting patches centered at body's joint locations and using an LSTM network for analyzing sequence. proposed methods have also focused on Dance videos.

II. RELATED WORKS

Computer vision has been increasingly applied to various domains, including dance analysis. Techniques such as motion capture and skeleton tracking have been used to study

dance movements, providing insights into pose accuracy and fluidity. However, most existing methods are tailored to Western dance forms, leaving a gap in research focused on traditional dances like Udarata[1-5].

Neural networks, particularly deep learning models, have shown great promise in pose estimation tasks. Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks are commonly used to capture spatial and temporal features, respectively. LSTMs are particularly effective for dynamic pose analysis due to their ability to handle sequential data[7,11,12,14].

Mediapipe is an open-source framework developed by Google that facilitates the detection of body landmarks in real-time. It has been successfully used in various applications, including fitness tracking and virtual reality. This framework's robustness and accuracy make it a suitable choice for our dynamic pose detection methodology[16,17].

Previous research on dance pose validation has primarily concentrated on ballet and contemporary dance. These studies generally employ motion capture systems and machine learning algorithms to assess pose accuracy. However, there is a notable gap in research on traditional dance forms such as Udarata, underscoring the need for further exploration in this area.

TABLE 1: Findings and Authors

Author, Title	Findings	Limitations
Author-cao et al,2017 Title- OpenPose Realtime MultiPerson 2D Pose Estimation using Part Affinity Fields	This groundbreaking work introduced OpenPose, A system for real-time pose detection of human poses in images. It set a new standard in	It doesn't fully grasp the cultural nuances of traditional dance, which can lead to inaccuracies.
	pose estimation across various datasets.	

Author - Donahue, J., et al. Title-Long-Term Recurrent Convolutional Networks for Visual Recognition and Description	This paper showcased how LSTMs can effectively recognize complex actions over time, significantly improving action recognition rates.	The focus was primarily on broad action categories, making it less applicable to specific styles of traditional dance
Author- Kwon, H., et al. Title-Dance Pose Recognition Using Deep Learnin	This study combined CNNs and LSTMs to recognize dance poses, achieving promising results, especially for structured dance forms.	The dataset was limited to only a few dance styles, which may not translate well to the unique poses found in dance.
Author - Kim, J., et al. Title-Pose Estimation and Action Recognition in Dance Videos	This paper proposed a framework for pose estimation and action recognition specifically tailored for dance videos, achieving high accuracy	Its effectiveness depends heavily on the quality of the training dataset, which may not cover all dance styles

III. DATA PREPARATION

A. Static Pose (Mandiya)

For data preparation, we collected images of various Udarata dance poses, ensuring a diverse representation of both correct and incorrect forms. Each image was labeled accordingly and stored in a dataset. These labeled images serve as the basis for our static pose detection methodology, allowing us to compare new poses against the dataset Fig1, Fig 2 represent the correct and incorrect Mandiya dance poses



Figure 1:Correct Mandiya Poses



Figure 2:Incorrect Mandiya Poses

B. Dynamic Pose

Dynamic poses data was gathered by recording videos of dance video poses performed by experienced dancers. Correct measurements are taken to validate the correct and incorrect poses.

IV. MODEL DEVELOPMENT

We have collected data by tracking key landmarks of the body for both dynamic and static dance poses. This data will

be instrumental in the next phases of our project, where it will be used to train and evaluate models designed to recognize and validate dance poses

TABLE 2:key Landmarks for Pose Detection in Body

pose detection key landmarks of the body	
Left shoulder	left knee
Right shoulder	right knee
Left elbow	left ankle
Right elbow	right ankle
Left wrist	left thumb
Right wrist	right thumb
Left hip	left index
Right hip	right index

A. Static Pose(Mandiya)

Static pose detection methodology involves comparing new images to the labeled dataset using image similarity metrics. Correct pose angles and heights are given and it comparatively compares the new pose with the given data. And it validates using given data.

Basically uses the computer vision algorithm to detect and validate the poses whether it is right or wrong

B. Dynamic Pose

For dynamic pose detection, we utilized Mediapipe to extract body landmarks from each video frame. These landmarks were used to create sequences representing the dance movements. These videos were then divided into individual frames, and each frame was annotated with body landmark coordinates using Mediapipe.

Frames were further labeled as correct or incorrect based on the pose accuracy, resulting in a comprehensive dataset for dynamic pose analysis. Correct video poses labeled as 1 and wrong video poses labeled as 0. The annotated frames were compiled into a CSV file, capturing the x and y coordinates of body landmarks. Each row in the CSV file represents a frame, labeled as either right or wrong. This structured data format facilitates the training of our LSTM model, enabling it to learn from both spatial and temporal patterns in the dance sequences. An LSTM model was then trained on these sequences, learning to classify them as correct or incorrect. The LSTM's ability to capture temporal dependencies makes it ideal for analyzing dynamic dance poses. Fig.3 illustrate the process flowchart.

C. LSTM Model

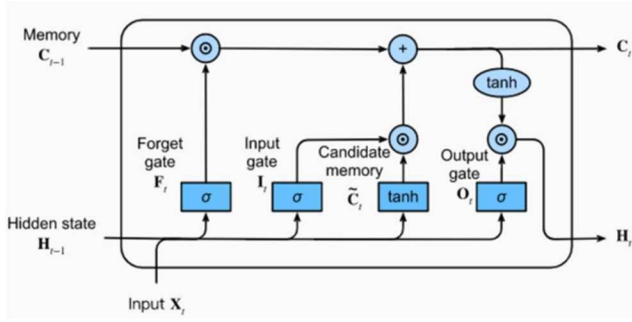


Figure 4:LSTM Model

Fig. 4 indicates the LSTM Model and the Long ShortTerm Memory LSTM model is a special type of neural network that's great at working with sequences, like video or time-series data, where understanding patterns over time really matters. Unlike standard neural networks, LSTMs are built to "remember" information over longer periods, which helps them pick up on long-term dependencies. For Udarata dance pose detection, LSTMs are particularly valuable. They can observe sequences of dance moves and recognize the flow of movements, making it easier to spot correct poses and highlight mistakes. By analyzing each frame in the sequence, the LSTM can detect patterns that match the dance style, offering real-time feedback on how well each step aligns with the traditional form

V. RESULTS

A. Static Pose (Mandiya)

Fig.5 represents the validation of a static dance pose, specifically the "Mandiya" pose, using a pose detection system. The pose detection system identifies and marks key landmarks on the body, such as the legs, shoulders, elbows, wrists, and thumbs. The system uses these measurements to validate the pose against a predefined set of correct angles and landmark positions for the "Mandiya" pose.



Figure 5: Correct Pose Validation

In this case, the system has labeled the pose as "CORRECT" based on the calculated angles and positions, indicating that the dancer's form matches the expected standard for this pose

This form of validation is crucial in determining whether the dancer is executing the pose correctly according to traditional dance standards. It allows for objective analysis and correction, which is particularly useful in training and performance settings



figure 6:Wrong Pose Validation

In this case, fig.6 the system has labeled the pose as "INCORRECT" based on the calculated angles and positions, indicating that the dancer's form unmatched the expected standard for this pose

B. Dynamic pose

The structured data was used to train a Long Short-Term Memory (LSTM) model. LSTMs are well-suited for this task due to their ability to capture and learn from temporal dependencies in sequential data, making them ideal for analyzing dynamic poses in dance sequences.

The model was trained to classify the sequences of frames as correct or incorrect based on the patterns it learned from the spatial and temporal data.

The Dynamic pose detection methodology achieved high accuracy, effectively identifying correct and incorrect poses from new videos. Performance metrics such as precision, recall, and F1-score demonstrated the robustness of our approach. Comparisons with baseline methods further validated the superiority of our technique.

TABLE 3: Accuracy Table

Dynamic pose	accuracy	F1 score	Recall score	precision
Goda saramba 1	98%	98.5%	98.5%	98.5%
Goda Saramba 2	92%	92%	91.5%	92%
Goda Saramba 3	92%	91%	92%	92%

high performance across different sequences, with accuracy and precision values ranging between 91% and 98.5%.

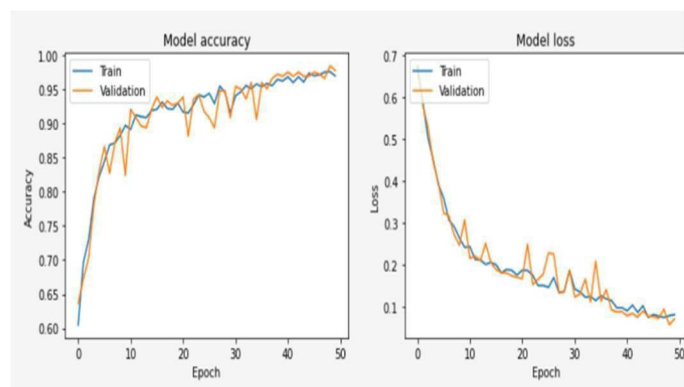


Figure 7:Accuracy Graph for Goda Saramba 1

The table summarizing the accuracy, F1 score, recall, and precision for different dynamic poses indicates consistently. The accuracy and loss graphs Fig.7 illustrate the performance of the LSTM model during the training and validation phases. These graphs demonstrate that the model achieved high accuracy with minimal loss across the training epochs, indicating robust learning.

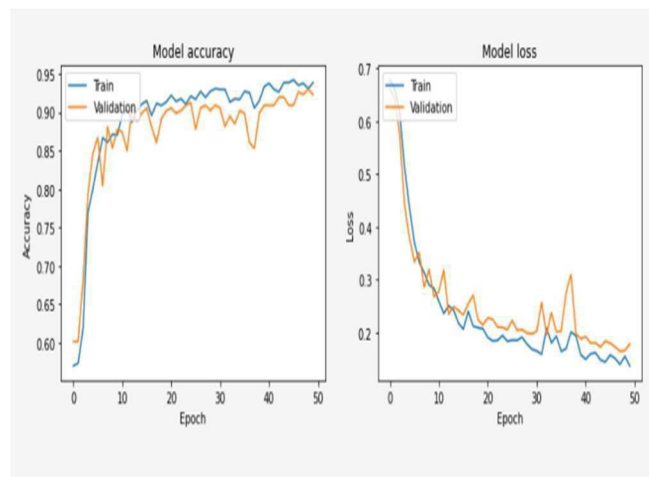


Figure 9:Accuracy Table for the Goda Saramba 2

The model shows fig.9 rapid improvement in both accuracy and loss during the early epochs. Training accuracy surpasses 95%, while validation accuracy stabilizes above 90%, indicating good generalization. Training loss decreases sharply toward zero, and validation loss also declines, though with more fluctuations. This suggests the model is effectively learning and minimizing errors throughout the training process.

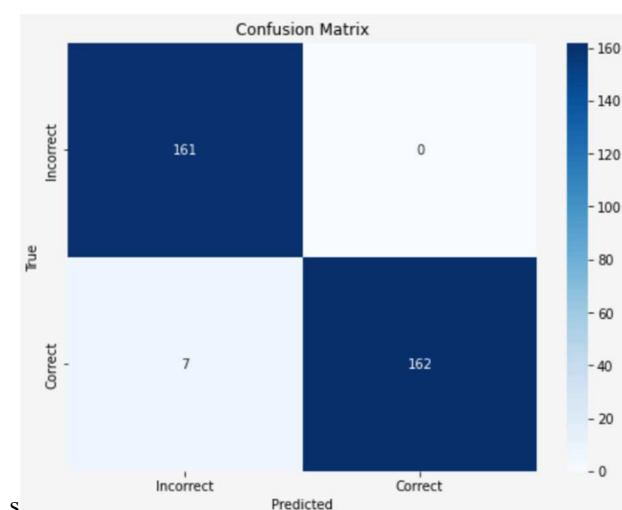


Figure 8:Confusion matrix of Goda Saramba 1

The confusion matrix fig.8 demonstrates the model's high accuracy in classifying dance poses, with 162 correct poses and 161 incorrect poses accurately identified. It shows a very low error rate, with only 7 correct poses being misclassified as incorrect, and no incorrect poses being wrongly classified as correct. This indicates that the model is highly effective and reliable for distinguishing between correct and incorrect dance poses, with minimal false negatives and no false positives.

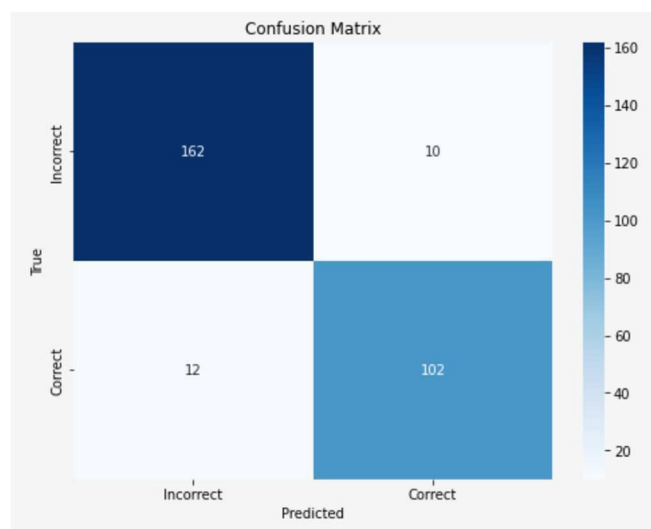


Figure 10:Confusion Matrix for Goda Saramba 2

Confusion matrix fig.10 shows that the model performs well, accurately identifying 162 'Incorrect' instances and 102 'Correct' instances. However, there are some misclassifications, with 10 'Incorrect' instances wrongly predicted as 'Correct' and 12 'Correct' instances wrongly predicted as 'Incorrect.' Overall, the model is more accurate at identifying 'Incorrect' instances than 'Correct' ones.

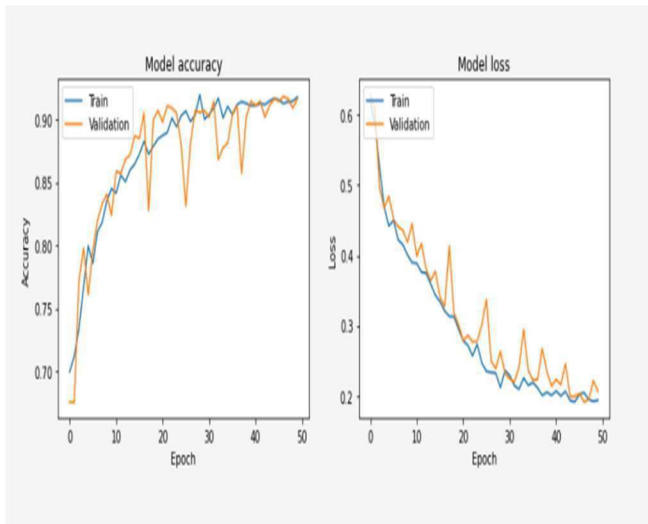


Figure 11: Accuracy Graph for Goda Saramba 3

The model's performance, fig.11 as depicted in the accuracy and loss graphs, shows a strong learning process over 50 epochs. The training accuracy steadily increases from around 70% to over 95%, while the validation accuracy follows a similar trend, stabilizing between 90-95%, indicating that the model generalizes well to unseen data. Concurrently, the training loss decreases significantly, approaching near zero by the end, with the validation loss also showing a downward trend despite some fluctuations. This consistent decrease in loss and the increase in accuracy across both training and validation datasets suggest that the model is effectively minimizing errors and improving its predictive accuracy over time.

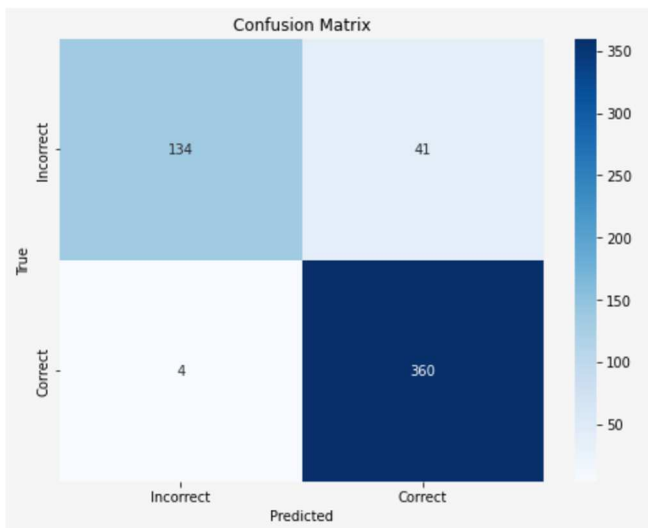


Figure 12: Confusion matrix for Goda Saramba 3

The fig.12 model accurately classified 360 instances as 'Correct,' reflecting strong performance in identifying this class. The model correctly identified 134 instances as 'Incorrect,' showing good performance in rejecting incorrect classifications. The model mistakenly classified 41 instances as 'Correct' when they were actually 'Incorrect,' indicating a moderate error rate in this regard. There were only 4 instances where the model incorrectly classified 'Correct' instances as

'Incorrect,' demonstrating high accuracy in identifying correct cases.

This analysis highlights the effectiveness of the LSTM model in accurately detecting and validating dynamic dance poses, as demonstrated by the high performance metrics and the clear separation of correct and incorrect classifications in the confusion matrices. The LSTM model exhibited strong performance in classifying dynamic dance sequences. Confusion matrices and other evaluation metrics indicated high accuracy and low error rates. The model's ability to generalize across different dancers and sequences underscores its effectiveness in dynamic pose detection. Our proposed methodologies outperformed traditional techniques in both static and dynamic pose detection. The use of advanced computer vision and deep learning models provided a significant advantage, enabling more accurate and reliable pose validation.

VI. CONCLUSION AND FUTURE WORKS

This research successfully demonstrated the application of computer vision and deep learning techniques to Udarata dance pose detection using LSTM. Both static and dynamic methodologies showed high accuracy and robustness, validating their effectiveness. The findings have significant implications for Udarata dance instruction and preservation. The automated detection and validation tools developed in this study can assist teachers in providing objective feedback, thereby improving the quality of dance education. Despite the promising results, our research has limitations. The dataset, though comprehensive, is limited in size and diversity. Additionally, the static pose detection methodology relies heavily on the quality of input images, which can vary in real-world scenarios. Future research could focus on expanding the dataset to include a wider range of poses and dancers. Enhancements to the model, such as incorporating more advanced neural network architectures or integrating additional contextual information, could further improve the accuracy and robustness of pose detection.

VII. REFERENCES

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